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A Contributions

Laura Ruis: project proposal and leadership, dataset development, code writing, human experiment, manual error analysis, paper writing and editing.

Akbir Khan: code writing, model evaluations, human experiment, paper writing and editing.

Stella Biderman: model evaluations, compute usage, paper writing and editing, advisor.

Sara Hooker: compute usage, paper writing and editing, advisor.

Tim Rocktäschel: paper writing and editing, advisor.

Edward Grefenstette: initial idea, manual error analysis, paper writing and editing, advisor.

B Reproducibility Statement

We share all the data, human annotations, code used for the evaluations, and the raw results in the supplementary material. Additionally, in Appendix K.5 we estimate the variance due to stochasticity in the API’s of OpenAI and Cohere. Of course, if either OpenAI or Cohere decides to change the models behind the API, the results might look different. We publish the exact date and time each API was queried for the results in Appendix L. Finally, in Appendix K.2 we estimate the variance over the prompt order of the in-context examples. The compute used for the experiments is detailed in Appendix M. The remaining experiments were done with API credits received as a research grant from OpenAI and Cohere.

C Ethics Statement

In this work, we conduct a study with human subjects (see Appendix H for details). To get matched with participants, we used the platform Prolific. Prolific complies with ethical standards according to UK law (e.g. complying with the GDPR). We compensated participants with a UK living wage at 15 GBP an hour, which is 6 GBP an hour more than Prolific recommends at 9 GBP per hour.

Implicature is an aspect of pragmatics, and pragmatic language impairments are universal in Autism Spectrum Disorder (ASD) (American Psychiatric Association, 2013). Difficulties in understanding scalar implicatures are claimed to be present in people with ASD (Volden, 2017), although the nature of the relation has proven hard to establish and has recently been debated (Katsos et al., 2011; Schaecken et al., 2018). For the purposes of this work, whether or not implicature understanding relates to ASD is not important. We took the following steps to make sure no sensitive data is collected or published. The human annotations we obtain are anonymous, related to a participant only by their Prolific ID for the purposes of compensation. In publishing the human annotations, we will not publish the Prolific ID of participants or anything else related to the participants. Additionally, we did not collect or request any personal or demographic characteristics of the participants apart from that they are all native English speakers.

Additionally, in this work we run a lot of compute-intensive experiments. We publish the estimated emissions per experiment in Appendix M. The total amount of GPU hours is estimated at maximally 966. How this is broken down per experiment can be seen in Appendix M.